Higher Eckmann-Hilton Arguments

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The Eckmann-Hilton Argument

Theorem (Eckmann-Hilton)

Let $(M, *, e)$ *and* $(M, \circ, e)$ *be unital magmas on the same set sharing a unit. Suppose the interchange law is satisfied for all* $a, b, c, d \in M$*:*

$$(a * b) \circ (c * d) = (a \circ c) * (b \circ d)$$

Then $* = \circ$ *and both are commutative, that is* $a * b = a \circ b = b * a = b \circ a$ *for all* $a, b \in M$

The Eckmann Hilton Argument

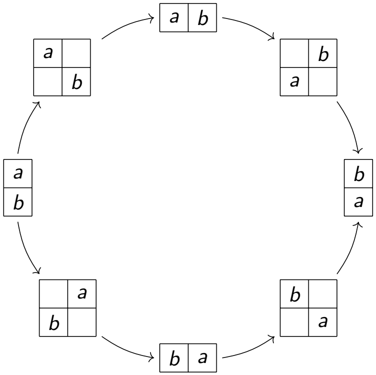
Proof.

$$
\begin{aligned}
& a * b \\
=& (a \circ e) * (e \circ b) && \text{(unitality)} \\
=& (a * e) \circ (e * b) && \text{(interchange)} \\
=& a \circ b && \text{(unitality)} \\
=& (e * a) \circ (b * e) && \text{(unitality)} \\
=& (e \circ b) * (a \circ e) && \text{(interchange)} \\
=& b * a && \text{(unitality)}
\end{aligned}
$$

$\square$

The Eckmann-Hilton Clock

We can represent this graphically, writing $\begin{array}{|c|}\hline a \\ \hline b \\ \hline \end{array}$ for $a * b$, $\begin{array}{|c|c|}\hline a & b \\ \hline \end{array}$ for $a \circ b$, and $\square$ for e. The interchange law is automatic from this notation.



Higher Categories

Higher categories provide a framework for proof-relevant versions of this argument.

Definition (Sketch)

An ω-**category** consists of:

- a set of 0-cells $x, y, \ldots$
- for any n-cells x, y, a set of $(n+1)$-cells $f : x \to y$ between them. (We say x is the n-dimensional source of f, and y is the n-dimensional target)
- for any n-cell x, an identity cell $\mathrm{e}_x : x \to x$
- for any $k \in \{0, \ldots, n-1\}$ and two n-cells f, g such that the k-dimensional target of f is the k-dimensional source of g, a k-composite cell $f *_k g$.
- coherence cells expressing isomorphisms, like $\delta_x : \mathrm{e}_x *_0 \mathrm{e}_x \cong \mathrm{e}_x$

Examples

$$x \xrightarrow{f} y \qquad y \xrightarrow{h} z \quad \rightsquigarrow \quad x \xrightarrow{f *_0 h} z$$

$$\begin{array}{c}
\begin{tikzcd} x \arrow[r, bend left, "f"] \arrow[r, bend right, "g"'] \arrow[r, phantom, "\Downarrow a"] & y \end{tikzcd} \\
\begin{tikzcd} x \arrow[r, bend left, "g"] \arrow[r, bend right, "h"'] \arrow[r, phantom, "\Downarrow b"] & y \end{tikzcd}
\end{array}
\quad \rightsquigarrow \quad
\begin{tikzcd} x \arrow[rr, bend left=60, "f"] \arrow[rr, bend right=60, "h"'] \arrow[rr, phantom, "\Downarrow a *_1 b"] & & y \end{tikzcd}$$

$$\begin{tikzcd} x \arrow[r, bend left, "f"] \arrow[r, bend right, "g"'] \arrow[r, phantom, "\Downarrow a"] & y \end{tikzcd}
\qquad
\begin{tikzcd} y \arrow[r, bend left, "h"] \arrow[r, bend right, "k"'] \arrow[r, phantom, "\Downarrow c"] & z \end{tikzcd}
\quad \rightsquigarrow \quad
\begin{tikzcd} x \arrow[rr, bend left=60, "f *_0 h"] \arrow[rr, bend right=60, "g *_0 k"'] \arrow[rr, phantom, "\Downarrow a *_0 b"] & & z \end{tikzcd}$$

$$\begin{tikzcd} & x \arrow[dl, equal, "\mathrm{e}_x"'] \arrow[dr, equal, "\mathrm{e}_x"] \arrow[d, phantom, "\Downarrow \delta_x"] & \\ x \arrow[rr, equal, "\mathrm{e}_x"'] & {} & y \end{tikzcd}$$

Proof-Relevant Eckmann-Hilton

2-cells whose source/target is $e_x \to e_x$ "weakly" satisfy the hypotheses of the Eckmann-Hilton argument:

- $*_1$ gives a (weakly) unital magma:

$$
\begin{tikzcd}
x \arrow[r, bend left=60, ""{name=U, below}] \arrow[r, ""{name=M}] \arrow[r, bend right=60, ""{name=D, above}] \arrow[Rightarrow, no head, from=U, to=M, "e_x^2"] \arrow[Rightarrow, from=M, to=D, "a"] & x
\end{tikzcd}
\cong
\begin{tikzcd}
x \arrow[r, bend left, ""{name=U, below}] \arrow[r, bend right, ""{name=D, above}] \arrow[Rightarrow, from=U, to=D, "a"] & x
\end{tikzcd}
$$

- $*_0$ can be "padded" to give a weakly unital magma:

$$
\begin{tikzcd}
x \arrow[r, bend left] \arrow[r, bend right] \arrow[r, phantom, "\Downarrow e_x^2"] \arrow[rr, bend left=60] \arrow[rr, bend right=60] & x \arrow[u, phantom, "\Downarrow \delta_x^{-1}"] \arrow[d, phantom, "\Downarrow \delta_x"] \arrow[r, bend left] \arrow[r, bend right] \arrow[r, phantom, "\Downarrow a"] & x
\end{tikzcd}
\cong
\begin{tikzcd}
x \arrow[r, bend left, ""{name=U, below}] \arrow[r, bend right, ""{name=D, above}] \arrow[Rightarrow, from=U, to=D, "a"] & x
\end{tikzcd}
$$

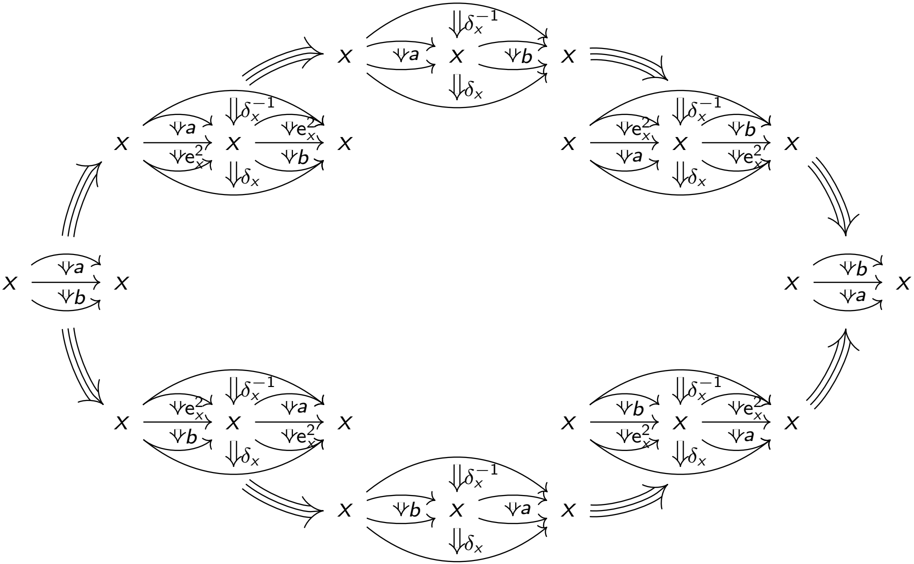
Proof-Relevant Eckmann-Hilton

- $*_0$ and $*_1$ satisfy a (weak) interchange law:

$$
\begin{tikzcd}
x \arrow[r, bend left=60] \arrow[r] \arrow[r, bend right=60] \arrow[r, phantom, shift left=3, "\Downarrow a"] \arrow[r, phantom, shift right=3, "\Downarrow b"] & x
\end{tikzcd}
*_0
\begin{tikzcd}
x \arrow[r, bend left=60] \arrow[r] \arrow[r, bend right=60] \arrow[r, phantom, shift left=3, "\Downarrow c"] \arrow[r, phantom, shift right=3, "\Downarrow d"] & x
\end{tikzcd}
\cong
\begin{array}{c}
\begin{tikzcd}
x \arrow[r, bend left=50] \arrow[r, bend right=50] \arrow[r, phantom, "\Downarrow a"] & x \arrow[r, bend left=50] \arrow[r, bend right=50] \arrow[r, phantom, "\Downarrow c"] & x
\end{tikzcd} \\
*_1 \\
\begin{tikzcd}
x \arrow[r, bend left=50] \arrow[r, bend right=50] \arrow[r, phantom, "\Downarrow b"] & x \arrow[r, bend left=50] \arrow[r, bend right=50] \arrow[r, phantom, "\Downarrow d"] & x
\end{tikzcd}
\end{array}
$$

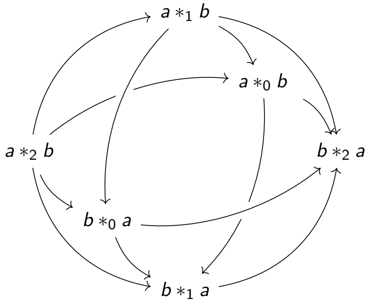
Thus we can construct a weak Eckmann-Hilton argument!

Proof-Relevant Eckmann-Hilton



Higher Dimensions?

Higher-dimensional cells satisfy even more Eckmann-Hilton relations. In dimension 3, the above Eckmann-Hilton clock can be expanded to the **Eckmann-Hilton sphere:**



We have produced the first explicit proofs of all these relations in all dimensions! They are formalized in the type theory `CaTT`, which can then be exported into homotopy type theory.

References

[1] Thibaut Benjamin. 'A type theoretic approach to weak ω-categories and related higher structures'. Thèse de doctorat. Institut Polytechnique de Paris, 2020.

[2] Thibaut Benjamin et al. *Higher Eckmann-Hilton Arguments in Type Theory*. 2025. arXiv: 2501.16465 [math.CT].

[3] Beno Eckmann and Peter J. Hilton. 'Structure maps in group theory'. In: *Fundamenta Mathematicae* 50 (1961), pp. 207–221. DOI: 10.4064/fm-50-2-207-221.